

- Monit can act automatically when an error occurs. For instance, if your Apache server is using too many resources, it can automatically restart Apache and send you an alert message.

Installation

Monit is meant for Linux and UNIX systems, so all you need to do is open your command line and type in the commands given below. It's just that simple.

For Debian-based Linux distros, the command is:

```
$ sudo apt-get install monit
```

For RPM-based systems, such as Fedora, CentOS and more, you need to work a little harder. Let's first install some prerequisites, as follows:

```
$ yum -y install openssl* gcc
$ yum install pam-devel
$ yum install openssl-devel
```

Now you need to download the latest version of Monit from <http://mmonit.com/monit>

```
$ cd /usr/src/
$ wget http://mmonit.com/monit/dist/monit-5.5.tar.gz
```

Extract the *tar* file now, as follows:

```
$ tar zxvf monit-5.5.tar.gz
```

Compile from the source:

```
$ cd monit-5.5/
$ ./configure
$ make && make install
```

Configuring and running Monit

By now, your system is up with Monit, and ready to retrieve and display system information. Along with the other great features, Monit possesses a great Web interface on Port 2812. All you need to do now is edit the Monit config file *monitrc* located at */etc/monit/*, according to your information needs. Let's get started.

First, you need to either edit the file or change file permissions by command, as the root user:

```
$ chmod 0700 /etc/monit/monitrc
```

The very first thing you probably want to configure is enabling the Web interface. Just add the following commands:

```
set httpd port 2812 and
    use address <server.example.net>
    allow <username>:<password>
```



Figure: Monitoring with the web interface

With the above configuration, you can browse to the location *server.example.net:2812* from your browser, with *username* and *password* as the login credentials.

Next, let's configure Monit for email alerts, as follows:

```
set mailserver <smtp.example.net>
username <smtp-username> password <smtp-password>
set mail-format { from: server@example.net}
```

In the above configuration, *smtp.example.net* is an SMTP server, whereas the *smtp-username* and *smtp-password* are your login credentials. Much of all this is self-explanatory.

Monitoring the system

To monitor the general health and resources of the system, you need to append the lines below to the *monitrc* file:

```
check system <server.example.net>
    if loadavg(1min) > 4 then alert
    if loadavg(5min) > 2 then alert
    if memory usage > 75% then alert
    if swap usage > 25% then alert
    if cpu usage (user) > 70% then alert
    if cpu usage (system) > 30% then alert
    if cpu usage (wait) > 20% then alert
```

Monitoring Apache Web server

To monitor Apache, you need to find the Apache process section in the *monitrc* file and uncomment it to produce the following result. Don't forget to take care of the *pid* file.

```
check process apache2 with pidfile /var/run/apache2.pid
start program = "/etc/init.d/apache2 start" with timeout 60
seconds
stop program = "/etc/init.d/apache2 stop"
if cpu > 60% for 2 cycles then alert
if cpu > 80% for 5 cycles then restart
if children > 250 then restart
if loadavg(5min) greater than 10 for 8 cycles then stop
if 3 restarts within 5 cycles then timeout
group server
```